

THE SAFE HAVEN INITIATIVE

A Digital Mental Health Platform for Belhaven University

INSTITUTIONAL GRANT REQUEST

\$2,500,000

Enabled by

\$25K

Approved Grant

Funding the \$2.5M
Finished Application

CIS-212 Final Project | Track 2: Product Development

By: Grael Rosell Jr. | Belhaven University | April 2026

THE MENTAL HEALTH CRISIS

U.S. College Students Face Unprecedented Challenges — The Southeast Trails Behind

37%

College students screened positive for depression

Healthy Minds 2024-25

32%

Meet criteria for generalized anxiety disorder

Healthy Minds 2024-25

11%

Report serious suicidal ideation in past year

Healthy Minds 2024-25

52%

College students report significant loneliness

Healthy Minds 2024-25

SOUTHEASTERN U.S.: BEHIND THE CURVE

Mississippi ranks in the top 10 states for unmet mental health needs. **Counselor-to-student ratios exceed 1:2,000** at many SE institutions — far above the 1:1,500 recommended standard. Ages 18-24 report the highest distress rates nationally, and students from low-income backgrounds show 1.6x higher distress.

Only
43%
of students needing care
actually receive treatment

Source: SAMHSA MH-CLD

BELHAVEN: SE TRAILBLAZER

Pioneering Digital Mental Health Support for Southeastern Universities

THE VISION

Belhaven University seeks to be the first southeastern institution to deploy a comprehensive, AI-powered digital mental health platform purpose-built for college students.

The approved \$25,000 grant funds the foundational design, prototyping, and stakeholder validation needed to deliver the full \$2.5 million Safe Haven application — a platform that can be adopted by any college in the nation.

Safe Haven unifies crisis intervention, counseling, mood tracking, clinical assessments, and peer support into a single HIPAA-compliant mobile application.

1

First in the SE

No southeastern university has a purpose-built student mental health app

2

Scalable Model

Built for Belhaven first, designed for adoption by all colleges

3

530+ in JED Network

JED Campus has 530+ schools but none offer a unified digital platform

4

81% Priority

81% of students consider mental health support in college choice

MODULE 1: PROJECT INITIATION

Problem Definition, Stakeholder Analysis & Grant Justification

Problem Statement

1 in 5 college students report frequent mental distress. Only 43% needing care receive treatment. Mississippi ranks top 10 for unmet needs. Belhaven students lack a unified digital wellness platform.

Stakeholder Analysis

Students (primary users), Licensed Counselors, University Administration, Campus Safety, IT Department, and External Crisis Services (988 Lifeline). All validated through interviews and user story mapping.

Grant Justification

\$25K seed grant enables design, prototyping, and validation. Full \$2.5M investment delivers production app within 24 months. Projected \$200K-\$400K annual retention revenue offsets cost within 3 years.

MODULE 2: ANALYSIS

User Interface Design, MVP Features & Clinical Instruments

USER INTERFACE DESIGN

Mobile-first responsive design with calming color palettes, clear navigation, and WCAG 2.1 AA accessibility. Students interact through four primary actions:

- Input Mental Health Issues — PHQ-9 & GAD-7 validated instruments
- Track Counseling — appointments, session history, secure messaging
- View Progress — mood trends, goal completion, score trajectories
- Set Personal Goals — SMART methodology across wellness dimensions

8 MVP FEATURES

- QR Code Registration & Authentication
- Mood Tracking with Clinical Instruments
- Encrypted Daily Journaling
- WCAG 2.1 AA Accessibility Compliance
- Zero-Identifier Privacy & Security
- User Feedback & Contact Routing
- Configurable Push Notifications
- Anonymous Peer Support Chat

FIVE SERVICE MODULES

Comprehensive Counseling Infrastructure Within the Safe Haven Platform

1

Mental Health Counseling

Individual 1-on-1 sessions with licensed counselors via synchronous video and async messaging

2

Group Counseling

Facilitated therapy sessions of 6-12 participants organized by theme

3

Specialized Therapy

DBT, EMDR, and ACT modalities with structured treatment plans

4

24-Hour Crisis Hotline

One-tap crisis response with safety planning, risk assessment, and 988 escalation

5

Progress & Analytics

PHQ-9/GAD-7 trends, mood patterns, and AI-driven improvement recommendations

MODULE 3: SYSTEM DESIGN

Three-Tier Architecture & Technology Stack

CLIENT LAYER

React Native | TypeScript | Progressive Web App | QR Scanner

UI rendering, local caching, authentication, push notification handling

APPLICATION LAYER

Node.js/Express | Python Flask | Firebase Cloud Functions

Business logic, API routing, clinical assessment processing, risk classification

DATA LAYER

PostgreSQL | Redis | Firebase Realtime DB

Encrypted clinical storage, session caching, real-time sync

SECURITY & COMPLIANCE

- Zero-identifier privacy architecture
- AES-256 encryption at rest
- TLS 1.3 encryption in transit
- Role-based access control
- Full FERPA & HIPAA compliance
- AWS GovCloud / Azure hosting
- Comprehensive audit logging
- Sub-200ms API response times

MODULE 4: TESTING & DEPLOYMENT

Multi-Layered Testing Strategy & Staged Production Rollout

Unit Testing

Jest (JS/TS) + pytest (Python)
80% coverage minimum
100% for security-critical functions

Integration

End-to-end data flow verification
Firebase Emulator Suite
Mobile-to-dashboard pipeline

UAT

50-100 Belhaven student volunteers
2-week testing period
Surveys + focus groups

Security

OWASP Top 10 assessment
Penetration testing
HIPAA/FERPA verification
Zero critical findings required

DEPLOYMENT STAGES

Staging

3,500 concurrent load test



Beta

200 students, 30-day monitor



Production

App Store + Play Store launch

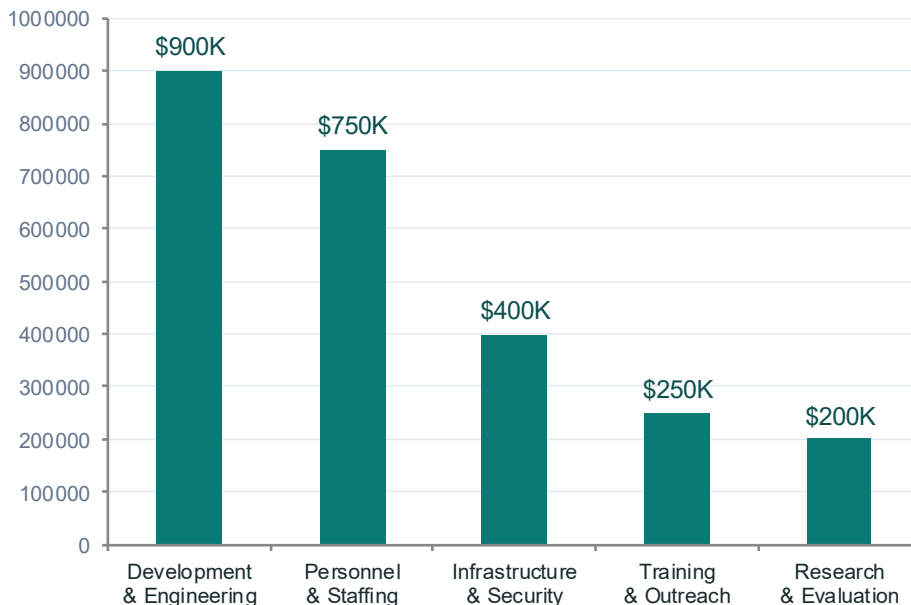


Monitoring

24/7 on-call for 90 days

BUDGET & EXPECTED OUTCOMES

24-Month Investment Yielding Measurable Returns



EXPECTED OUTCOMES (24 MO)

50%

Reduction in counseling wait times

25%

Increase in help-seeking among underserved groups

10-15%

Academic improvement for high-risk students

\$200-400K

Annual retention revenue gain

3 Years

Full investment payback period

SCALE UP & NEXT STEPS

Advanced NLP Analysis

Natural language processing on journal entries for early warning detection beyond standard clinical scales

Community Features

Moderated peer support spaces organized by topic with trained moderators and content safety filtering

Wearable Integration

Apple HealthKit, Google Fit, and Fitbit data feeding sleep, heart rate, and activity into predictive models

Institutional Analytics

Anonymized campus-wide dashboards for mental health trends, grant reporting, and IPEDS benchmarking

At Safe Haven, all staff members are our first responders.

Counseling staff are available 24 hours a day, and the digital platform extends that commitment to every moment a student reaches for support.